# STK4021 – Mandatory assignment, Fall 2026

Bayesian inversion of seismic data

NAME – STK4021

15 October 2026

## Self-report

*To be written by the student: how the work proceeded, what was difficult, time-consuming or fun, what was relatively easy, collaboration and sources used (including AI tools).*

# Part 1

## 1a)

$m$ and $\varepsilon$ are independent and Gaussian, and $d$ is linear in them, so $(m, d)$ is jointly Gaussian with

$$\begin{aligned} E(d) &= A\mu_m, \\ \mathrm{Cov}(d) &= A\Sigma_m A^\top + \sigma_\varepsilon^2 I \;=:\; \Sigma_d, \\ \mathrm{Cov}(d, m) &= \mathrm{Cov}(Am + \varepsilon, m) = A\Sigma_m, \end{aligned}$$

so

$$\begin{pmatrix} m \\ d \end{pmatrix} \sim N\left( \begin{pmatrix} \mu_m \\ A\mu_m \end{pmatrix}, \begin{pmatrix} \Sigma_m & \Sigma_m A^\top \\ A\Sigma_m & \Sigma_d \end{pmatrix} \right).$$

Conditioning on $d$ gives $m \mid d \sim N(\mu_{m|d}, \Sigma_{m|d})$ with

$$\begin{aligned} \mu_{m|d} &= \mu_m + \Sigma_m A^\top \Sigma_d^{-1}(d - A\mu_m), \\ \Sigma_{m|d} &= \Sigma_m - \Sigma_m A^\top \Sigma_d^{-1} A\Sigma_m. \end{aligned}$$

## 1b)

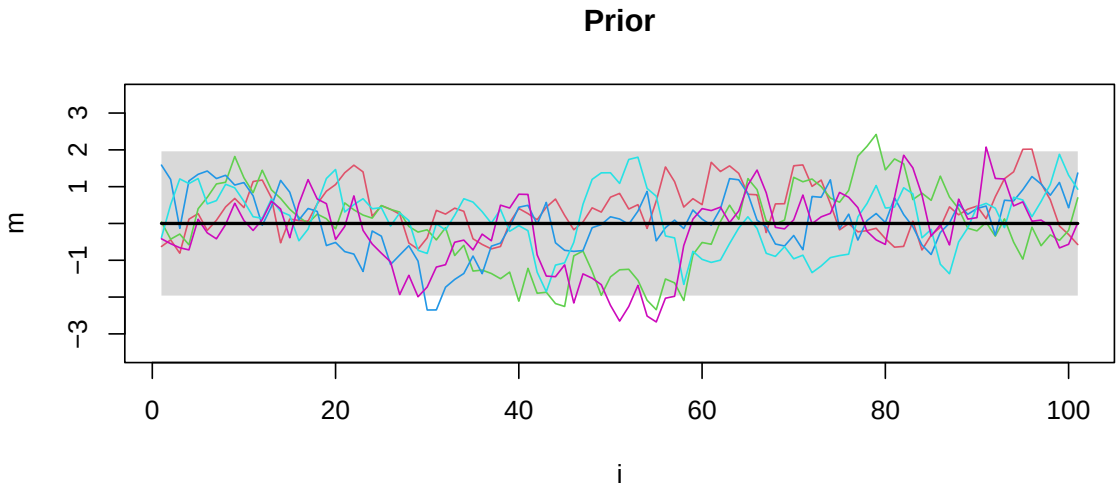



Figure 1: Prior expectation (black), 95% credibility interval (grey) and five simulated profiles.

**1c)**

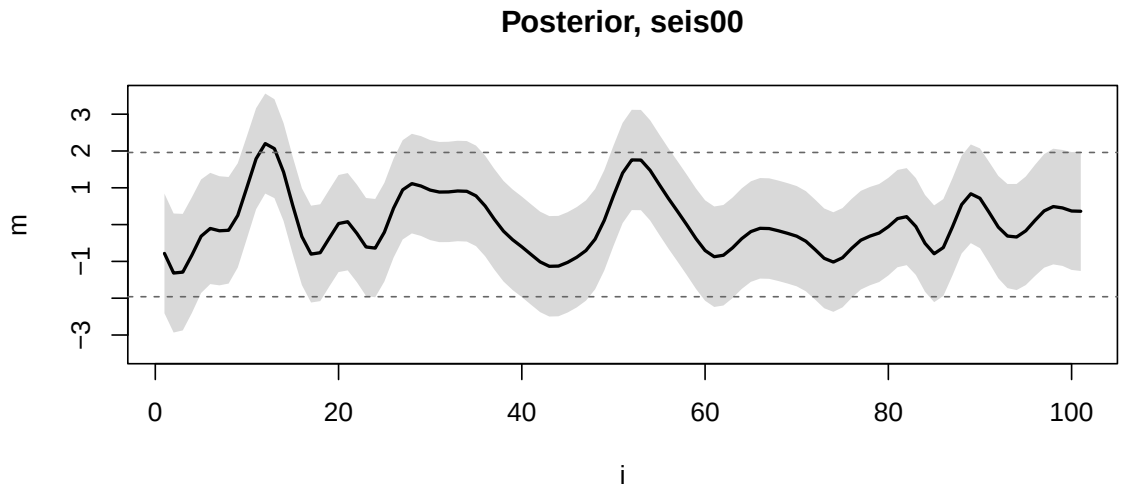



Figure 2: Posterior expectation (black) and 95% credibility interval (grey) for seis00. Dashed: prior interval.

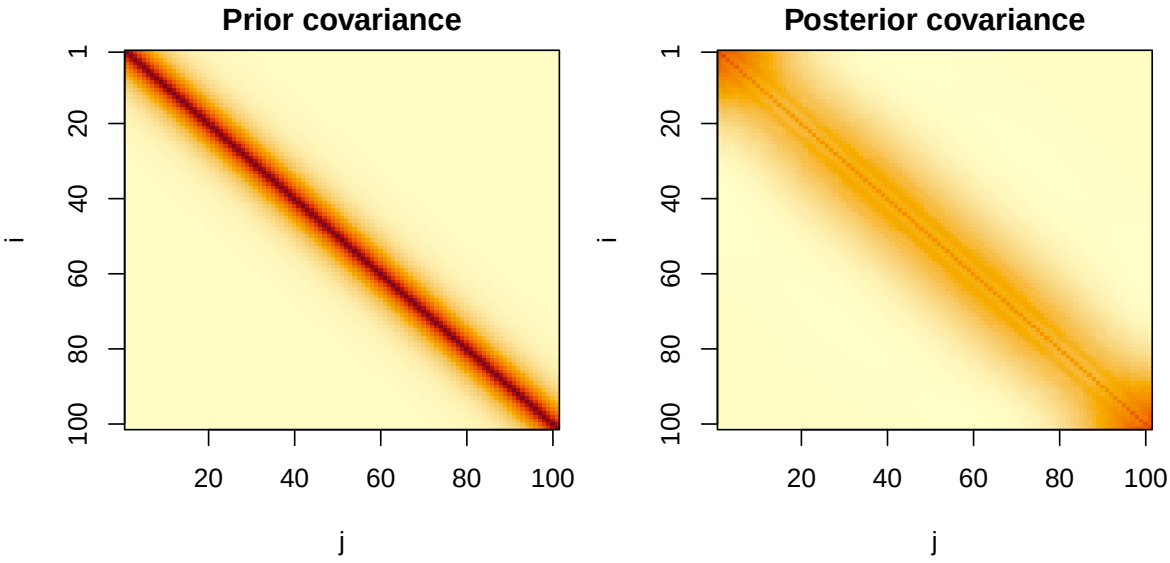



Figure 3: Prior (left) and posterior (right) covariance matrices, same colour scale.

**1d)**

Table 1: Posterior mean, median and sd of $V$ ($B = 100$).

| mean | median | sd |
|---|---|---|
| 8.76 | 5.48 | 8.68 |

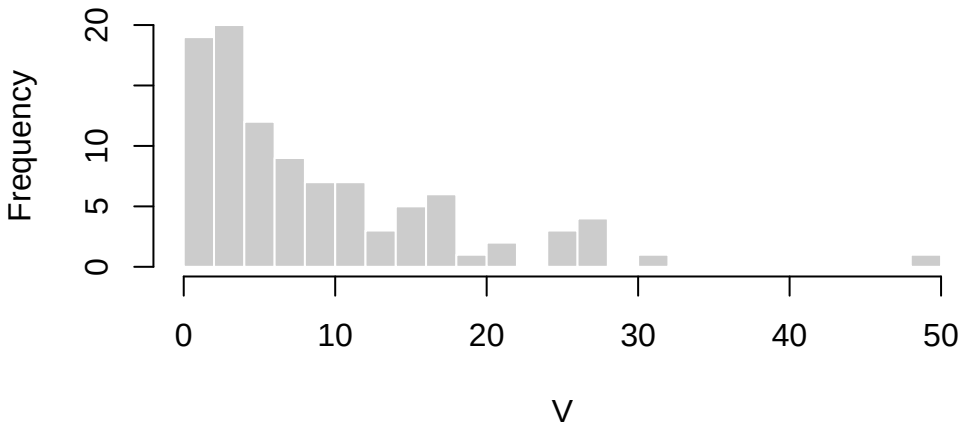



Figure 4: Histogram of 100 posterior samples of $V$.

## 1e)

We use the 10th and 90th smallest of the 100 samples: $\widehat{P}_{10} = 0.92$ and $\widehat{P}_{90} = 20.64$.

The number of samples below $P_{10}$ is $N \sim \text{Bin}(100, 0.1)$, and $P_{10}$ lies between the $r$-th and $s$-th smallest sample when $r \leq N \leq s-1$:

$$P\big(V_{(r)} \leq P_{10} < V_{(s)}\big) = \sum_{k=r}^{s-1} \binom{100}{k} 0.1^k 0.9^{100-k}.$$

With $r = 5$ and $s = 17$ this is 0.956, so a 95% interval for $P_{10}$ is $[0.19,\ 1.82]$. This is wide, so 100 samples is not enough.

**1f)**

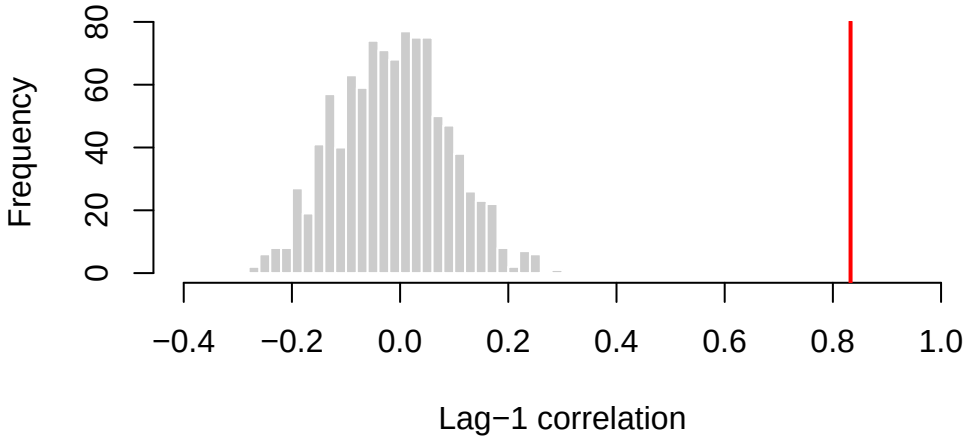



Figure 5: Lag-1 correlation for 1000 permutations of seis00 (histogram) and for the observed data (red line).

The observed lag-1 correlation is 0.83, larger than all 1000 permuted values ($p < 0.001$). The data are therefore not exchangeable.

# Part 2

## 2a)

Only the prior mean depends on $(T, B)$, so from 1a

$$p(d \mid T, B) = N\big(d;\ A\mu_m^{T,B},\ \Sigma_d\big), \qquad \Sigma_d = A\Sigma_m A^\top + \sigma_\varepsilon^2 I,$$

and by Bayes' rule, for $(T, B) \in \mathcal{G}$,

$$p(T, B \mid d) = \frac{p(d \mid T, B)\, p(T, B)}{\sum_{(T', B') \in \mathcal{G}} p(d \mid T', B')\, p(T', B')} \ \propto\ \exp\Big\{ -\tfrac{1}{2}(d - A\mu_m^{T,B})^\top \Sigma_d^{-1} (d - A\mu_m^{T,B}) \Big\},$$

since the uniform prior and the normalising constant do not depend on $(T, B)$.

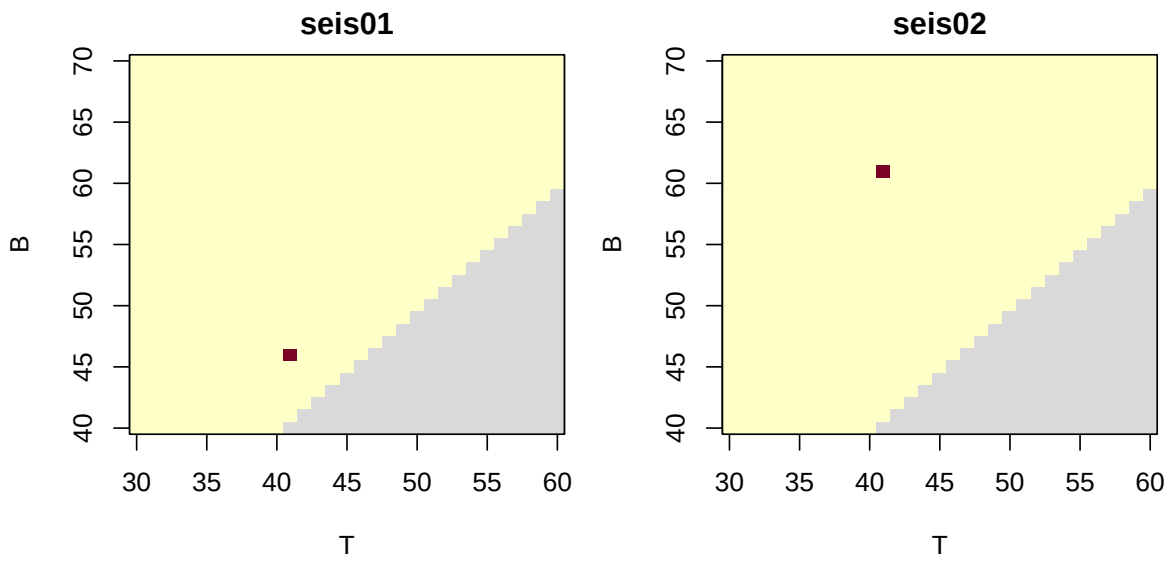



Figure 6: Joint posterior $p(T, B \mid d)$ for seis01 (left) and seis02 (right). Grey: inadmissible pairs.

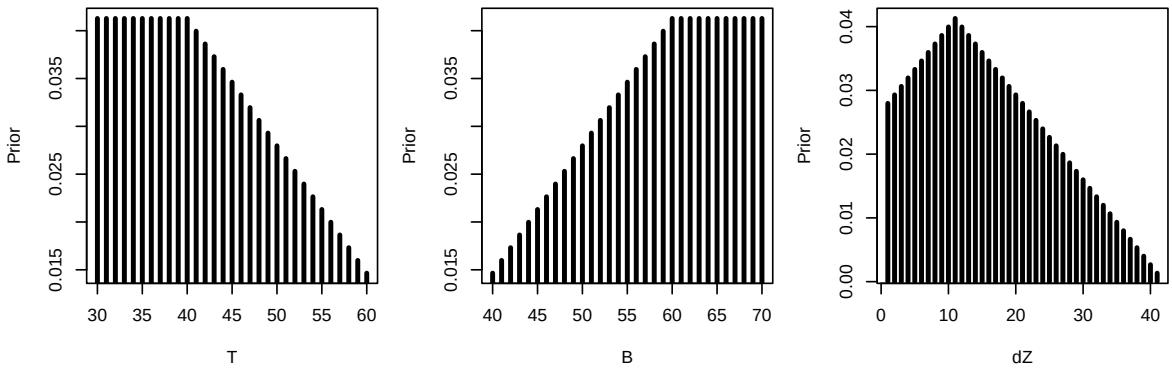



Figure 7: Marginal prior of top $T$, base $B$ and thickness $\Delta Z$ (uniform prior on $\mathcal{G}$).

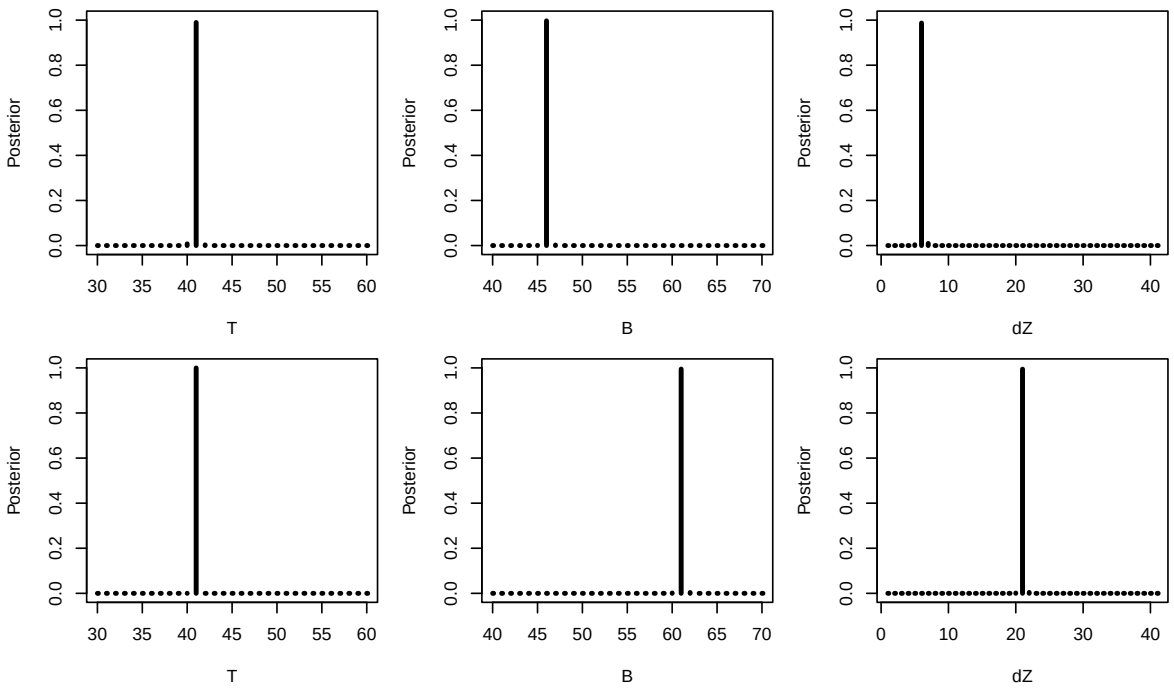



Figure 8: Marginal posterior of $T$, $B$ and $\Delta Z$. Top row: seis01, bottom row: seis02.

The prior marginals are not uniform, since the number of possible partners varies. The posterior puts almost all mass on $(41, 46)$ for seis01 (probability 0.987) and $(41, 61)$ for seis02 (probability 0.995), i.e. thickness 6 and 21. The data determine the zone almost exactly.

## 2b)

If all 752 models (751 zones and no zone) got the same prior probability, $P(\text{no zone})$ would be only $1/752$, just because there are many ways to place a zone. The prior for the zero case should instead reflect how likely we think it is that a zone exists. We set $P(\text{no zone}) = 1/2$ and spread the rest uniformly over $\mathcal{G}$:

$$P(\text{no zone}) = \tfrac{1}{2}, \qquad p(T, B) = \tfrac{1}{2} \cdot \tfrac{1}{|\mathcal{G}|},\ (T, B) \in \mathcal{G},$$

$$P(\text{no zone} \mid d) = \frac{\frac{1}{2}\, p(d \mid \text{no zone})}{\frac{1}{2}\, p(d \mid \text{no zone}) + \frac{1}{2}\, \frac{1}{|\mathcal{G}|} \sum_{\mathcal{G}} p(d \mid T, B)}.$$

This gives $P(\text{no zone} \mid d) = 2.1 \times 10^{-8}$ for seis01 and $3.4 \times 10^{-13}$ for seis02.

## 2c)

Let $a_1 =$ invest and $a_0 =$ do not invest, with loss equal to minus the profit:

$$L(a_1, V) = -(aV - b), \qquad L(a_0, V) = 0.$$

The Bayes decision minimises the expected posterior loss. Since $E\{L(a_1, V) \mid d\} = -(aE(V \mid d) - b)$ and $E\{L(a_0, V) \mid d\} = 0$, the rule is

$$\text{invest} \iff E(V \mid d) > \frac{b}{a} = 10.$$

We find $E(V \mid d)$ by simulation: draw $(T, B)$ (or no zone) from the posterior in 2b, draw $m$ from the posterior in 1a with prior mean $\mu_m^{T,B}$, and compute $V$.

Table 2: Posterior summaries of $V$ and the profit $aV - b$ (5000 simulations).

| | E(V\|d) | E(profit\|d) | P(profit<0\|d) | P10 | P50 | P90 |
|---|---|---|---|---|---|---|
| seis01 | 31.17 | 21.17 | 0 | 11.66 | 19.80 | 32.75 |
| seis02 | 89.56 | 79.56 | 0 | 60.69 | 78.71 | 99.45 |

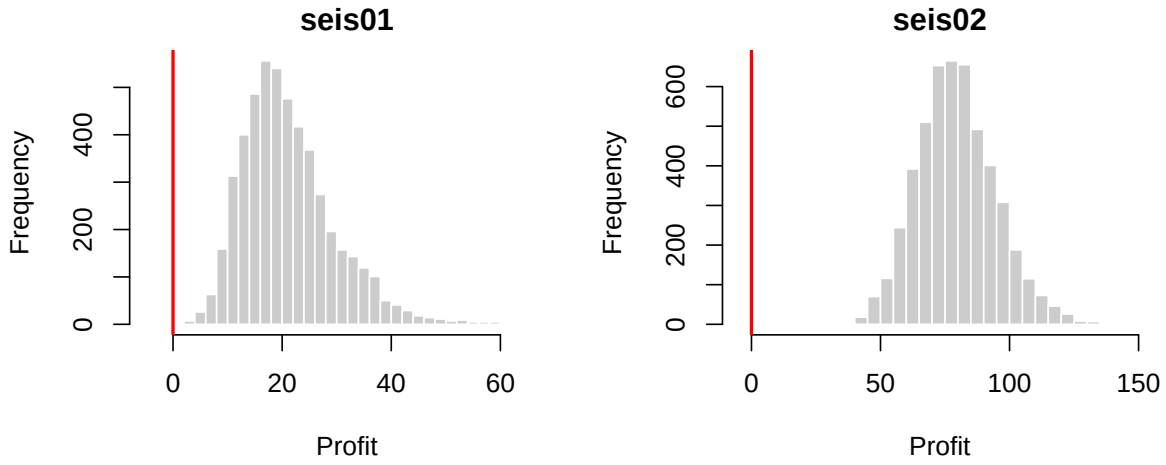



Figure 9: Posterior distribution of the profit $V - 10$. Red line: break-even.

We recommend investing in both cases, since $E(V \mid d)$ is far above 10 and none of the 5000 simulations gave a loss. The uncertainty is shown by the profit histograms and the percentiles in the table.

# Part 3

## 3a)

We write $m = \mu 1 - \Delta\mu\, 1_{[T,B]} + x$ with $x \sim N(0, \Sigma_m)$ independent of $\mu \sim N(\mu_0, \tau_\mu^2)$. By double expectation and total variance,

$$E(m) = E\{E(m \mid \mu)\} = \mu_0 1 - \Delta\mu\, 1_{[T,B]},$$
$$\mathrm{Cov}(m) = E\{\mathrm{Cov}(m \mid \mu)\} + \mathrm{Cov}\{E(m \mid \mu)\} = \Sigma_m + \tau_\mu^2 11^\top,$$

and therefore

$$E(d) = AE(m), \qquad \mathrm{Cov}(d) = A\Sigma_m A^\top + \tau_\mu^2 (A1)(A1)^\top + \sigma_\varepsilon^2 I.$$

With $\mu_0 = 0$ the mean is unchanged. The rows of $A$ sum to zero, so $A1 = 0$ and the extra term disappears. So $p(d \mid T, B)$ and $p(T, B \mid d)$ are the same as in 2a.

Recomputing for seis01 gives the same posterior (largest difference $2.6 \times 10^{-16}$). This is not surprising, since the data say nothing about the overall level of $m$.

## 3b)

With $\sigma_\varepsilon^2 = \nu\sigma_m^2$ and $R = \Sigma_m/\sigma_m^2$ (the correlation matrix), $\mathrm{Cov}(d) = \sigma_m^2 C$ with $C = ARA^\top + \nu I$, so for a given $(T, B)$

$$p(d \mid \sigma_m^2, T, B) \propto (\sigma_m^2)^{-n/2} \exp\Big\{ - \frac{Q_{T,B}}{2\sigma_m^2} \Big\}, \qquad Q_{T,B} = (d - A\mu_m^{T,B})^\top C^{-1} (d - A\mu_m^{T,B}).$$

The conjugate prior is $\sigma_m^2 \sim \text{IG}(\alpha, \beta)$, and then

$$p(\sigma_m^2 \mid d, T, B) \propto (\sigma_m^2)^{-(\alpha+n/2)-1} \exp\Big\{ -\frac{\beta + Q_{T,B}/2}{\sigma_m^2} \Big\} \;\Rightarrow\; \sigma_m^2 \mid d, T, B \sim \text{IG}\big(\alpha + \tfrac{n}{2},\; \beta + \tfrac{1}{2}Q_{T,B}\big).$$

Since $(T, B)$ is unknown, $p(\sigma_m^2 \mid d) = \sum_{\mathcal{G}} p(T, B \mid d)\, \text{IG}(\sigma_m^2; \alpha + \frac{n}{2}, \beta + \frac{1}{2}Q_{T,B})$, with $p(T, B \mid d)$ from 3c. We use $\alpha = 2$, $\beta = 1$ (prior mean 1) and $\nu = 0.04$.

Table 3: Posterior mean and 95% credibility interval for $\sigma_m^2$.

| | posterior mean | 2.5% | 97.5% |
|---|---|---|---|
| seis01 | 1.27 | 0.97 | 1.66 |
| seis02 | 1.05 | 0.79 | 1.37 |

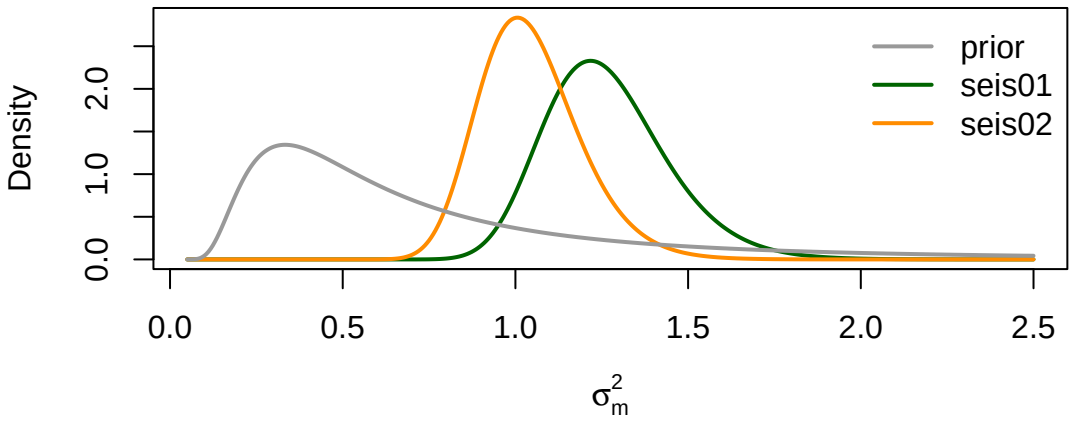



Figure 10: Prior IG(2,1) (grey) and posterior densities of $\sigma_m^2$ for seis01 and seis02.

### 3c)

We account for the uncertainty by integrating $\sigma_m^2$ out:

$$p(d \mid T, B) = \int p(d \mid \sigma_m^2, T, B)\, p(\sigma_m^2)\, d\sigma_m^2 \;\propto\; \Big(\beta + \tfrac{1}{2}Q_{T,B}\Big)^{-(\alpha+n/2)}$$

(a multivariate Student-$t$ density), so $p(T, B \mid d) \propto (\beta + \frac{1}{2}Q_{T,B})^{-(\alpha+n/2)}$, compared with $\propto \exp(-\frac{1}{2}Q_{T,B})$ for fixed $\sigma_m^2 = 1$.

Table 4: Posterior of $(T, B)$ with fixed $\sigma_m^2 = 1$ (columns 1–3) and with $\sigma_m^2$ integrated out (columns 4–6).

| | mode | prob. | E(dZ\|d) | mode | prob. | E(dZ\|d) |
|---|---|---|---|---|---|---|
| seis01 | (41,46) | 0.987 | 6.01 | (41,46) | 0.954 | 6.02 |

| | mode | prob. | E(dZ\|d) | mode | prob. | E(dZ\|d) |
|---|---|---|---|---|---|---|
| seis02 | (41,61) | 0.995 | 21 | (41,61) | 0.991 | 21 |

The most likely pair is the same, but the posterior is slightly wider: the probability of the mode drops from 0.987 to 0.954 (seis01) and from 0.995 to 0.991 (seis02).

## Appendix: R code

```
knitr::opts_chunk$set(echo = FALSE, message = FALSE, warning = FALSE,
                      fig.align = "center", fig.width = 7.5, fig.height = 3.6,
                      out.width = "95%", fig.pos = "H")
# ---- Data ----
A <- as.matrix(read.table("data/A.txt"))
dimnames(A) <- NULL
d00 <- scan("data/seis00.txt", quiet = TRUE)
d01 <- scan("data/seis01.txt", quiet = TRUE)
d02 <- scan("data/seis02.txt", quiet = TRUE)
n <- nrow(A)
idx <- 1:n
# ---- Helper functions ----

# exponential covariance, S_ij = s2 * exp(-3|i-j|/l)
cov_exp <- function(n, s2, l) {
  s2 * exp(-3 * abs(outer(1:n, 1:n, "-")) / l)
}

# posterior of m given d (Gaussian conditioning, 1a)
post_m <- function(d, mu, S, s2e) {
  S_d <- A %*% S %*% t(A) + s2e * diag(n)
  K <- S %*% t(A) %*% solve(S_d)
  S_post <- S - K %*% A %*% S
  list(mean = drop(mu + K %*% (d - A %*% mu)),
       cov = (S_post + t(S_post)) / 2)
}

# k draws from N(mu, S), one per row
draw_mvn <- function(k, mu, S) {
  L <- t(chol(S + 1e-10 * diag(length(mu))))
  Z <- matrix(rnorm(length(mu) * k), length(mu), k)
  t(mu + L %*% Z)
}

# mean with 95% band, optionally some realisations on top
plot_band <- function(mean, sd, title, ylim, sims = NULL) {
  plot(idx, mean, type = "n", ylim = ylim, xlab = "i", ylab = "m", main = title)
  polygon(c(idx, rev(idx)), c(mean - 1.96 * sd, rev(mean + 1.96 * sd)),
          col = "grey85", border = NA)
  if (!is.null(sims)) matlines(idx, t(sims), lty = 1, col = 2:6)
  lines(idx, mean, lwd = 2)
}
# ---- 1b ----
mu0 <- 0; s2m <- 1; l <- 20; s2e <- 0.04
mu_prior <- rep(mu0, n)
S_prior <- cov_exp(n, s2m, l)

set.seed(1)
prior_draws <- draw_mvn(5, mu_prior, S_prior)
plot_band(mu_prior, sqrt(diag(S_prior)), "Prior", c(-3.5, 3.5), prior_draws)
# ---- 1c ----
```

```r
post00 <- post_m(d00, mu_prior, S_prior, s2e)
plot_band(post00$mean, sqrt(diag(post00$cov)), "Posterior, seis00", c(-3.5, 3.5))
abline(h = c(-1.96, 1.96), lty = 2, col = "grey40")
plot_cov <- function(S, title, zlim) {
  image(idx, idx, t(S)[, n:1], zlim = zlim, col = hcl.colors(64, "YlOrRd", rev = TRUE),
        xlab = "j", ylab = "i", main = title, axes = FALSE)
  axis(1)
  axis(2, at = n + 1 - c(1, 20, 40, 60, 80, 100), labels = c(1, 20, 40, 60, 80, 100))
  box()
}
zlim <- range(S_prior, post00$cov)
par(mfrow = c(1, 2), mar = c(4, 4, 2, 1))
plot_cov(S_prior, "Prior covariance", zlim)
plot_cov(post00$cov, "Posterior covariance", zlim)
# ---- 1d ----
cval <- -1
volume <- function(m) sum((cval - m) * (m < cval))

B <- 100
set.seed(2)
m_draws <- draw_mvn(B, post00$mean, post00$cov)
V <- apply(m_draws, 1, volume)

V_stats <- c(mean = mean(V), median = median(V), sd = sd(V))
knitr::kable(t(round(V_stats, 2)), caption = "Posterior mean, median and sd of $V$ ($B=100$).")
hist(V, breaks = 20, col = "grey80", border = "white", main = "", xlab = "V")
# ---- 1e ----
V_sorted <- sort(V)
P10 <- V_sorted[10]
P90 <- V_sorted[90]

# number below P10 is Bin(B, 0.1)
r <- qbinom(0.025, B, 0.1)
s <- qbinom(0.975, B, 0.1) + 1
cover <- pbinom(s - 1, B, 0.1) - pbinom(r - 1, B, 0.1)
# ---- 1f ----
lag1 <- function(x) cor(x[-1], x[-length(x)])

rho <- lag1(d00)
set.seed(4)
rho_perm <- replicate(1000, lag1(sample(d00)))

hist(rho_perm, breaks = 30, col = "grey80", border = "white", main = "",
     xlab = "Lag-1 correlation", xlim = c(-0.4, 1))
abline(v = rho, col = "red", lwd = 2)
# ---- 2a ----
dmu <- 5
pairs <- subset(expand.grid(T = 30:60, B = 40:70), T <= B)
pairs$dZ <- pairs$B - pairs$T + 1

# prior mean with a zone from T to B (T = NA means no zone)
zone_mean <- function(T, B) {
  m <- rep(mu0, n)
```

```
  if (!is.na(T)) m[T:B] <- mu0 - dmu
  m
}

S_d <- A %*% S_prior %*% t(A) + s2e * diag(n)
L_d <- t(chol(S_d))

# log p(d | T, B) without the constant
loglik <- function(d, mu, L = L_d) {
  e <- forwardsolve(L, d - A %*% mu)
  -0.5 * sum(e^2)
}

logsumexp <- function(x) max(x) + log(sum(exp(x - max(x))))

post_TB <- function(d, L = L_d) {
  ll <- sapply(1:nrow(pairs), function(k) loglik(d, zone_mean(pairs$T[k], pairs$B[k]), L))
  list(ll = ll, prob = exp(ll - logsumexp(ll)))
}

zone01 <- post_TB(d01)
zone02 <- post_TB(d02)
plot_heat <- function(prob, title) {
  M <- matrix(NA, 31, 31)
  M[cbind(pairs$T - 29, pairs$B - 39)] <- prob
  image(30:60, 40:70, M, col = hcl.colors(64, "YlOrRd", rev = TRUE),
        xlab = "T", ylab = "B", main = title)
  image(30:60, 40:70, ifelse(is.na(M), 1, NA), col = "grey85", add = TRUE)
  box()
}
par(mfrow = c(1, 2), mar = c(4, 4, 2, 1))
plot_heat(zone01$prob, "seis01")
plot_heat(zone02$prob, "seis02")
marginal <- function(w, x, vals) sapply(vals, function(v) sum(w[x == v]))
prior_TB <- rep(1 / nrow(pairs), nrow(pairs))

par(mfrow = c(1, 3), mar = c(4, 4, 1, 1))
plot(30:60, marginal(prior_TB, pairs$T, 30:60), type = "h", lwd = 3, xlab = "T", ylab = "Prior")
plot(40:70, marginal(prior_TB, pairs$B, 40:70), type = "h", lwd = 3, xlab = "B", ylab = "Prior")
plot(1:41, marginal(prior_TB, pairs$dZ, 1:41), type = "h", lwd = 3, xlab = "dZ", ylab = "Prior")
par(mfrow = c(2, 3), mar = c(4, 4, 1, 1))
for (z in list(zone01, zone02)) {
  plot(30:60, marginal(z$prob, pairs$T, 30:60), type = "h", lwd = 3, ylim = c(0, 1),
       xlab = "T", ylab = "Posterior")
  plot(40:70, marginal(z$prob, pairs$B, 40:70), type = "h", lwd = 3, ylim = c(0, 1),
       xlab = "B", ylab = "Posterior")
  plot(1:41, marginal(z$prob, pairs$dZ, 1:41), type = "h", lwd = 3, ylim = c(0, 1),
       xlab = "dZ", ylab = "Posterior")
}
best_pair <- function(prob) {
  k <- which.max(prob)
  c(T = pairs$T[k], B = pairs$B[k], dZ = pairs$dZ[k], prob = prob[k])
}
```

```
best <- rbind(seis01 = best_pair(zone01$prob), seis02 = best_pair(zone02$prob))
# ---- 2b ----
p0 <- 1 / 2
prob_nozone <- function(z, d) {
  l0 <- log(p0) + loglik(d, zone_mean(NA, NA))
  l1 <- log(1 - p0) + logsumexp(z$ll) - log(nrow(pairs))
  exp(l0 - logsumexp(c(l0, l1)))
}
p_nozone <- c(seis01 = prob_nozone(zone01, d01), seis02 = prob_nozone(zone02, d02))
# ---- 2c ----
a <- 1; b <- 10; nsim <- 5000
K <- S_prior %*% t(A) %*% solve(S_d)
L_post <- t(chol(post00$cov + 1e-10 * diag(n)))

sim_volume <- function(z, d, p_no) {
  w <- c(p_no, (1 - p_no) * z$prob)
  Ts <- c(NA, pairs$T); Bs <- c(NA, pairs$B)
  pick <- sample(length(w), nsim, replace = TRUE, prob = w)
  sapply(pick, function(k) {
    mu_k <- zone_mean(Ts[k], Bs[k])
    m <- mu_k + K %*% (d - A %*% mu_k) + L_post %*% rnorm(n)
    volume(m)
  })
}
set.seed(5)
V01 <- sim_volume(zone01, d01, p_nozone[1])
V02 <- sim_volume(zone02, d02, p_nozone[2])
profit <- list(seis01 = a * V01 - b, seis02 = a * V02 - b)

profit_tab <- t(sapply(profit, function(x)
  c(mean(x + b) / a, mean(x), mean(x < 0), quantile(x, c(0.1, 0.5, 0.9)))))
knitr::kable(profit_tab, digits = 2,
             col.names = c("E(V|d)", "E(profit|d)", "P(profit<0|d)", "P10", "P50", "P90"),
             caption = "Posterior summaries of $V$ and the profit $aV-b$ (5000 simulations).")
par(mfrow = c(1, 2), mar = c(4, 4, 2, 1))
for (nm in names(profit)) {
  hist(profit[[nm]], breaks = 30, col = "grey80", border = "white", main = nm,
       xlab = "Profit", xlim = range(-5, profit[[nm]]))
  abline(v = 0, col = "red", lwd = 2)
}
# ---- 3a ----
tau2 <- 0.25
S_prior_mu <- S_prior + tau2 * matrix(1, n, n)
S_d_mu <- A %*% S_prior_mu %*% t(A) + s2e * diag(n)
zone01_mu <- post_TB(d01, L = t(chol(S_d_mu)))
max_diff <- max(abs(zone01_mu$prob - zone01$prob))
# ---- 3b ----
alpha <- 2; beta <- 1; nu <- 0.04
C <- A %*% (S_prior / s2m) %*% t(A) + nu * diag(n)
L_C <- t(chol(C))
quad_form <- function(d) sapply(1:nrow(pairs), function(k) {
  e <- forwardsolve(L_C, d - A %*% zone_mean(pairs$T[k], pairs$B[k]))
  sum(e^2)
```

```
})

# p(T, B | d) with s2m integrated out (used in 3c)
post_TB_t <- function(Q) {
  lw <- -(alpha + n / 2) * log(beta + Q / 2)
  exp(lw - logsumexp(lw))
}
Q01 <- quad_form(d01); Q02 <- quad_form(d02)
zone01_t <- post_TB_t(Q01); zone02_t <- post_TB_t(Q02)

# draws from p(s2m | d), a mixture of inverse gammas
draw_s2m <- function(Q, w, k = 20000) {
  j <- sample(length(w), k, replace = TRUE, prob = w)
  1 / rgamma(k, shape = alpha + n / 2, rate = beta + Q[j] / 2)
}
set.seed(6)
s2m_01 <- draw_s2m(Q01, zone01_t); s2m_02 <- draw_s2m(Q02, zone02_t)
s2m_tab <- rbind(seis01 = c(mean(s2m_01), quantile(s2m_01, c(0.025, 0.975))),
                 seis02 = c(mean(s2m_02), quantile(s2m_02, c(0.025, 0.975))))
knitr::kable(s2m_tab, digits = 2, col.names = c("posterior mean", "2.5%", "97.5%"),
             caption = "Posterior mean and 95\\% credibility interval for $\\sigma_m^2$.")
dinvgamma <- function(x, a, b) exp(a * log(b) - lgamma(a) - (a + 1) * log(x) - b / x)
dens_s2m <- function(x, Q, w) sapply(x, function(v) sum(w * dinvgamma(v, alpha + n / 2, beta + Q / 2)))

x <- seq(0.05, 2.5, length.out = 400)
f01 <- dens_s2m(x, Q01, zone01_t)
f02 <- dens_s2m(x, Q02, zone02_t)
plot(x, f01, type = "l", lwd = 2, col = "darkgreen", ylim = c(0, max(f01, f02)),
     xlab = expression(sigma[m]^2), ylab = "Density")
lines(x, f02, lwd = 2, col = "darkorange")
lines(x, dinvgamma(x, alpha, beta), lwd = 2, col = "grey60")
legend("topright", c("prior", "seis01", "seis02"), lwd = 2,
       col = c("grey60", "darkgreen", "darkorange"), bty = "n")
# ---- 3c ----
compare <- function(p_fix, p_t) {
  i <- which.max(p_fix); j <- which.max(p_t)
  c(mode_fix = paste0("(", pairs$T[i], ",", pairs$B[i], ")"), prob_fix = round(p_fix[i], 3),
    dZ_fix = round(sum(p_fix * pairs$dZ), 2),
    mode_t = paste0("(", pairs$T[j], ",", pairs$B[j], ")"), prob_t = round(p_t[j], 3),
    dZ_t = round(sum(p_t * pairs$dZ), 2))
}
cmp_tab <- rbind(seis01 = compare(zone01$prob, zone01_t), seis02 = compare(zone02$prob, zone02_t))
knitr::kable(cmp_tab, col.names = c("mode", "prob.", "E(dZ|d)", "mode", "prob.", "E(dZ|d)"),
             caption = "Posterior of $(T,B)$ with fixed $\\sigma_m^2=1$ (columns 1--3) and with $\\sigma
```